

KW 13: Inflation & The Phillips Curve

Chapter 8 + Lecture 6

Blanchard - Macroeconomics, 7th Global Edition

Study Notes

LECTURE 6 CONTEXT: INFLATION

Learning Objectives

- Economy in the medium-run
- Importance of inflation
- Explaining inflation: Quantity theory and Phillips curve
- Extension of the IS-LM model to include the real interest rate and risk premium

Literature: Blanchard, Macroeconomics, Chapters 8, 23.2, 23.3.

From the Labor Market to Inflation

Lecture 5 introduced the medium-run: resource constraints (labor) limit potential output. A persistently higher demand only leads to higher prices. We now study the topic of inflation.

Motivation: Venezuela in the IS-LM Model

- In the IS-LM model, increasing the money supply decreases interest rates, stimulates the economy, and increases GDP
- Venezuela expanded its money supply excessively -- the bolivar fuerte became practically worthless, with over 1,000,000% inflation in 2018
- On August 20, 2018, the new bolivar soberano was introduced at an exchange rate of 1:100,000, but lost value just as quickly
- Real GDP declined by more than two-thirds between 2013 and 2020

Motivation: Venezuela in the IS-LM model

- ▶ In the IS-LM model, an increase in the money supply leads to decreasing interest rates, which in turn stimulates the economy and increases GDP.
- ▶ Venezuela has done this excessively — with limited success:
 - ▶ The money supply in Venezuela grew so extremely in 2017/2018 that the bolívar fuerte became practically worthless, with over 1,000,000% inflation in 2018.
 - ▶ On August 20, 2018, the new bolívar soberano was introduced at an exchange rate of 1:100,000, but lost value just as quickly.
 - ▶ Real GDP growth declined by more than two thirds between 2013 and 2020. More than half of transactions now take place in dollars. Cryptocurrencies like the Petro are also used in some cases.
 - ▶ Data sources: CEIC Data (M2 Money Supply, Real GDP Growth, CPI)

Figure: Venezuela - Excessive money supply growth led to hyperinflation and economic collapse

INFLATION: DEFINITION AND MEASUREMENT

Definition

The inflation rate measures the rate of change of the price level over time:

$$\pi_{2025} \equiv \frac{P_{2025} - P_{2024}}{P_{2024}}$$

Measurement

- **Consumer Price Index (CPI):** Measures inflation in consumption. How does the price of an average basket of goods change?
- **GDP deflator:** Ratio of nominal GDP to real GDP. GDP deflator = Nominal GDP / Real GDP
- CPI and GDP deflator differ in: industrial and government consumption, imports, and the weighting of goods

Problems with Measuring Inflation

- Quality improvements and new products make measurement difficult (Aghion et al., AER, 2019)
- 'Shrinkflation': Same price but smaller quantity -- not captured by standard CPI

Swiss CPI Basket 2025 (Weights)

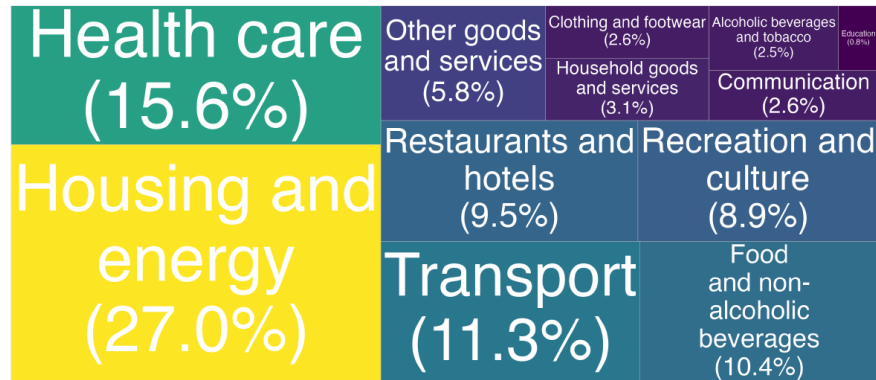
Category	Weight (%)
Housing and energy	27.0
Health care	15.6
Transport	11.3
Food and non-alcoholic beverages	10.4
Restaurants, hotels	9.5
Recreation and culture	8.9
Other goods and services	5.8
Clothing and footwear	3.6
Alcoholic beverages and tobacco	3.5
Household goods and services	3.1
Communication	2.6
Education	0.8 (est.)

For comparison: food had a weight of 21% in the 1980s. CPI basket weights differ across countries (e.g., food weight: Austria 12.1%, Latvia 26.8%), complicating international comparisons.

Inflation measurement in Switzerland

CPI basket 2025 – weights of main groups

Data: FSO



- ▶ For comparison: Food had a weight of 21% in the 1980s.
- ▶ Do all people consume goods in these proportions?

Figure: Swiss CPI Basket 2025 - Weights of main groups (Data: FSO)

Inflation Since 1970

- From the mid-1980s, inflation was a limited problem in most developed countries
- Temporary rise in inflation rates after Covid-19
- Swiss inflation has been persistently lower than in other countries

Inflation since 1970

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- Temporary rise in inflation rates after Covid-19:

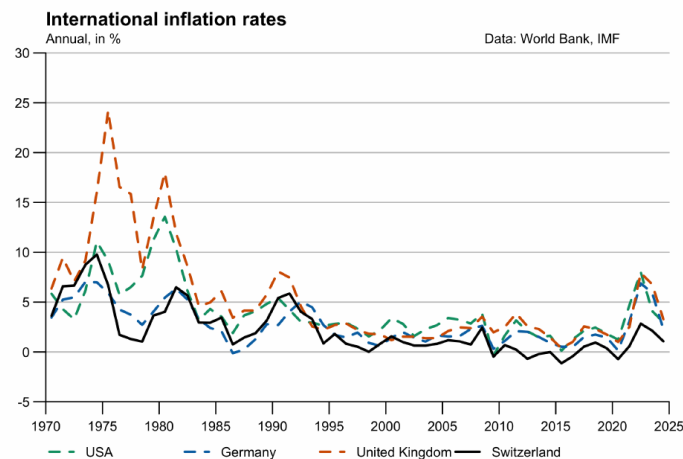


Figure: International inflation rates since 1970 - USA, Germany, UK, Switzerland (Data: World Bank, IMF)

Public Perception of Inflation

The correlation between actual inflation and expected price developments in Switzerland is high (0.8), suggesting the public has a reasonably good sense of inflation trends.

Does the public have a good sense of inflation?

- The correlation between inflation and expected price developments is high (0.8).

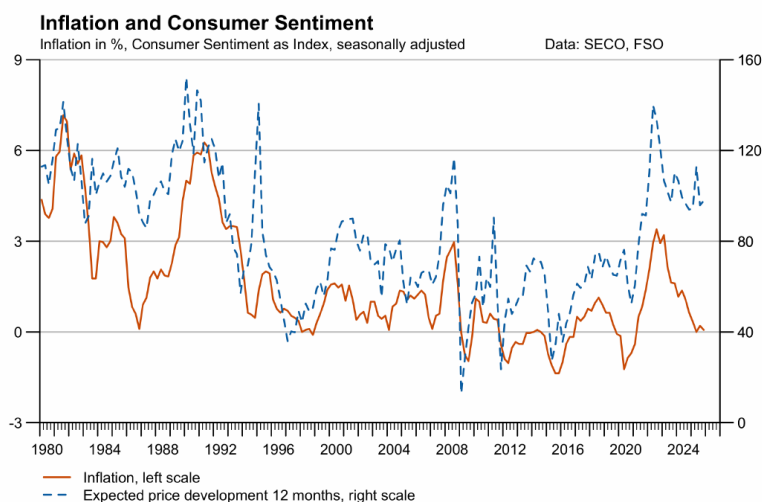


Figure: Swiss Inflation vs Consumer Sentiment (1980-2024, Data: SECO, FSO)

COSTS AND BENEFITS OF INFLATION

Costs of Inflation

- **Uncertainty:** High inflation is associated with uncertainty about future prices, making planning difficult
- **Redistribution:** From creditors to debtors (unexpected inflation erodes the real value of debt)
- **Nominal contracts:** Bracket creep (tax brackets not indexed), wages, pensions, and mortgages affected
- **Menu costs:** Costs of changing prices (reprinting menus, catalogs)
- **Shoe leather costs:** Costs of economizing on money holdings (more frequent bank trips)
- **Flight to real assets:** People convert money to material assets or stable foreign currencies (e.g., USD)
- **Political costs:** Inflation reduces re-election chances (behavioural economics: people 'do not like' inflation)
- Examples of hyperinflation: Germany 1922/23, Zimbabwe 2006-08

Benefits of Inflation

- **Central bank room for manoeuvre:** Inflation allows the central bank to set negative real interest rates, since nominal rates cannot become arbitrarily negative (zero lower bound)
- **Real wage adjustment:** Nominal wage cuts are hard to enforce; inflation leads to real wage reductions without explicit cuts
- **Seigniorage:** State revenues from the creation of money. This can be less distortionary than other taxes
- **Initial stimulus:** The initial consequences of higher inflation can be beneficial (short-run trade-off)

Why Not Target 0% Inflation?

If inflation is costly, why not aim for zero? Because moderate inflation (around 2%) gives central banks room to lower real interest rates during recessions (since nominal rates hit zero as a floor). Also, nominal wage rigidity means that real wage adjustments are easier with positive inflation. Stable inflation is better than no inflation -- see Avenir Suisse (2017).

QUANTITY THEORY OF MONEY

How Does Inflation Arise?

First explanation: the money supply increases too fast.

Money growth and inflation

- How does inflation arise?
- First explanation: The money supply increases (too fast).

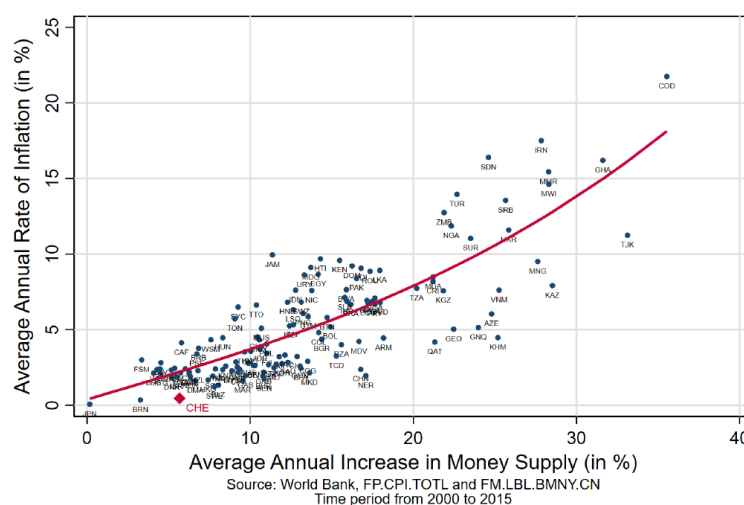


Figure: Cross-country data showing positive correlation between money supply growth and inflation (2000-2015)

Transaction Version

Milton Friedman (1963): 'Inflation is always and everywhere a monetary phenomenon.'

The equation of exchange:

$$M \cdot V = P \cdot T$$

- **M** = money supply
- **V** = velocity of money (how often each unit of money changes hands)
- **P** = price level
- **T** = number of transactions

Under two assumptions:

- Assumption 1: Money supply **M** has no impact on **T** (neutrality of money)
- Assumption 2: **V** is constant in the medium run ($\Delta V = 0$)

It follows that:

$$\% \Delta M = \% \Delta P$$

Money supply and price level are proportional to each other. Inflation is caused by growth of the money supply.

Income Version

Since the number of transactions T is difficult to measure, replace T with real GDP (Y):

$$M \cdot V = P \cdot Y$$

Swiss Data: Money Supply and Prices

Quantity theory of money — data from Switzerland

- How have money supply and prices developed in Switzerland?

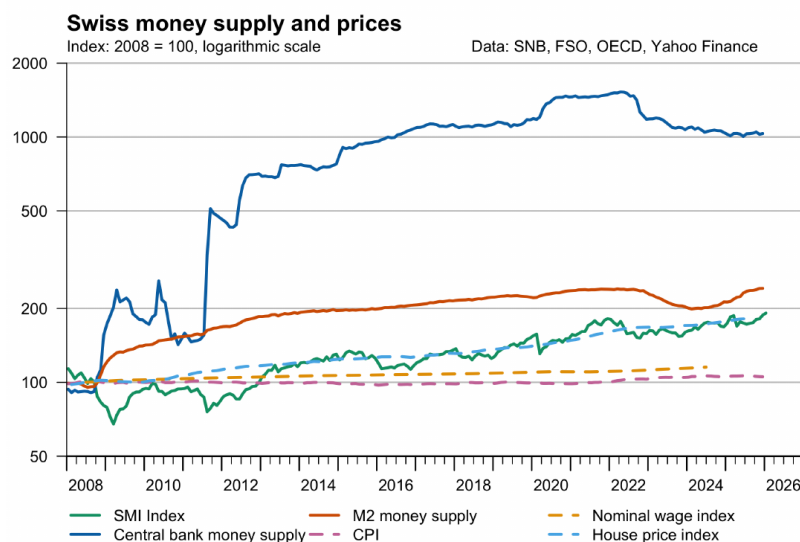


Figure: Swiss money supply ($M0$, $M2$) and prices (CPI, wages, SMI, house prices) since 2008 (Data: SNB, FSO)

Did QE Lead to Inflation?

- Central bank money supply ($M0$) increased massively due to quantitative easing, but the increase in $M2$ was much smaller
- Most of the money did not end up in the real economy -- the velocity of circulation V is not constant
- The inflation occurred primarily in financial markets (stock prices) and real estate, not measured by CPI
- Why was there no CPI inflation until 2021? Because asset price inflation (SMI, house prices) is not captured by the CPI

Quantity Theory: Useful but Limited

The quantity theory provides a useful long-run relationship between money growth and inflation. However, the assumption of constant velocity (V) is violated in the short and medium run, especially during periods of quantitative easing. The theory works

better for explaining hyperinflation episodes (Venezuela, Zimbabwe) than for moderate inflation in developed economies.

THE PHILLIPS CURVE

8-1 Historical Background

In 1958, A. W. Phillips plotted inflation against unemployment for the UK (1861-1957) and found a negative relationship. Samuelson and Solow (1960) replicated this for the US (1900-1960).

- When unemployment was low, inflation was high
- When unemployment was high, inflation was low or negative
- This was called the **Phillips curve**: it implied a trade-off between inflation and unemployment

The Phillips curve in practice

- The postulated relationship in Swiss time series data:

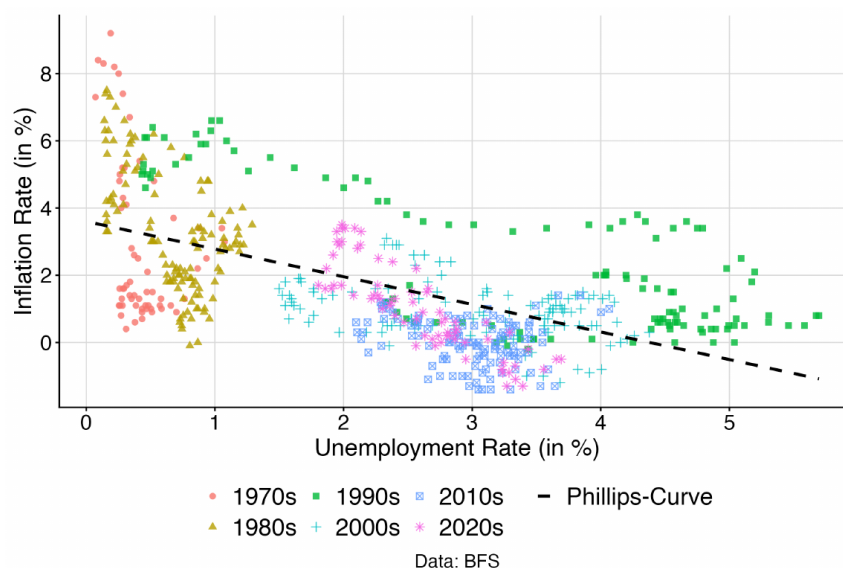


Figure: Phillips Curve in Swiss data by decade (1970s-2020s) - negative correlation between unemployment and inflation (Data: BFS)

The Phillips Curve -- Intuition (Lecture)

- How do companies react to additional demand?
 - > Overtime, hiring more staff, subcontracting, raising prices
 - > At high capacity utilization: primarily increase prices
- In a boom (low unemployment), wages tend to rise due to workers' stronger negotiating position
- With higher wages, companies pass on the increased wage costs as higher prices
- Time series data confirm: inflation (or wage growth) is higher in years with low unemployment

8-1 Deriving the Relation: Inflation, Expected Inflation, Unemployment

Starting from the wage-setting and price-setting equations from Chapter 7:

$$\text{Wage setting: } W = P^e \cdot F(u, z)$$

$$\text{Price setting: } P = (1 + \mu) \cdot W$$

Combining these (replace W from the first into the second):

$$P = P^e(1 + \mu)(1 - \alpha u + z) \quad (\text{Eq. 8.1})$$

where $F(u, z) = 1 - \alpha u + z$ is a specific functional form, and α captures the strength of unemployment's effect on wages.

By means of some transformations (dividing by P_{t-1} , using the approximation that cross-products of small terms vanish), we obtain:

$$\pi_t = \pi_t^e + (\mu + z) - \alpha u_t \quad (\text{Eq. 8.3})$$

The Core Relation (Eq. 8.3)

Inflation depends on: (1) expected inflation π^e -- higher expected inflation leads to higher actual inflation; (2) the markup μ and catch-all z -- higher markup or z increases inflation; (3) the unemployment rate u -- lower unemployment increases inflation. This equation is the foundation for understanding the Phillips curve and its mutations.

Effects of Each Variable

- **Expected inflation up** -> actual inflation up (workers demand higher wages anticipating price increases)
- **Markup μ or z up** -> inflation up (higher costs or wage demands push prices up)
- **Unemployment u down** -> inflation up (tighter labor market gives workers more bargaining power)

THE PHILLIPS CURVE AND ITS MUTATIONS

8-2 The Original Phillips Curve (Pre-1970)

When inflation is not persistent, it is reasonable for wage setters to assume expected inflation equals some constant $\bar{\pi}$:

$$\pi_t^e = \bar{\pi} \Rightarrow \pi_t = \bar{\pi} + (\mu + z) - \alpha u_t \quad (\text{Eq. 8.4})$$

- This gives a **negative relation between the level of inflation and unemployment**
- This is what Phillips, Samuelson, and Solow observed in the pre-1970 data
- It implied a **trade-off**: accept more inflation to get lower unemployment, or vice versa

Anchored Expectations (Lecture)

If the inflation rate is stable at a level π^* over several years and without a trend, rational individuals expect inflation of π^* for the next year:

$$\pi_t^e = \pi^*$$

- In this case, low unemployment leads to high inflation (with μ and z unchanged)
- It appears that there is a trade-off between inflation and unemployment
- Helmut Schmidt (1972): 'It seems to me that the German people can tolerate a 5% price increase more easily than a 5% unemployment rate'

The Apparent Trade-Off and Its Disappearance

In the 1960s, U.S. policy aimed at steadily decreasing unemployment. From 1961 to 1969, unemployment fell from 6.8% to 3.4% while inflation rose from 1.0% to 5.5%. The economy moved along the original Phillips curve.

Around 1970, the relation broke down. Both high inflation AND high unemployment appeared (stagflation). The original Phillips curve vanished.

8-2 Why the Phillips Curve Collapsed

Wage setters changed how they formed expectations. As inflation became persistent in the 1970s, people started basing expectations on last year's inflation:

$$\pi_t^e = (1 - \theta)\bar{\pi} + \theta\pi_{t-1} \quad (\text{Eq. 8.5})$$

- **theta = 0**: Expectations ignore past inflation. The original Phillips curve holds (Eq. 8.4)
- **0 < theta < 1**: Expectations partly depend on last year's inflation
- **theta = 1**: Expectations fully based on last year's inflation. The relation becomes:

$$\pi_t - \pi_{t-1} = (\mu + z) - \alpha u_t \quad (\text{Eq. 8.6})$$

Modified (Expectations-Augmented) Phillips Curve

When $\theta = 1$, the unemployment rate affects not the level of inflation, but the CHANGE in inflation. High unemployment leads to decreasing inflation; low unemployment leads to increasing inflation. This is also called the accelerationist Phillips curve (low u accelerates inflation).

The empirical counterpart for the US (1970-2014):

$$\pi_t - \pi_{t-1} = 3.0\% - 0.5 u_t \quad (\text{Eq. 8.7})$$

THE PHILLIPS CURVE AND THE NATURAL RATE

8-3 The Natural Rate of Unemployment

In the late 1960s, Friedman and Phelps argued that the inflation-unemployment trade-off was an illusion. They predicted it would disappear, and coined the term 'natural rate of unemployment'.

Deriving the Natural Rate

The natural rate u_n is the unemployment rate at which actual inflation equals expected inflation ($\pi_t = \pi_t^e$). Setting $\pi_t = \pi_t^e$ in Eq. 8.3:

$$0 = (\mu + z) - \alpha u_n \Rightarrow u_n = \frac{\mu + z}{\alpha} \quad (\text{Eq. 8.8})$$

- The higher the markup μ , the higher the natural rate
- The higher z (unemployment benefits, employment protection, etc.), the higher the natural rate
- The higher α (sensitivity of wages to unemployment), the lower the natural rate

Rewriting the Phillips Curve

Substituting $u_n = (\mu + z)/\alpha$ back into the Phillips curve:

$$\pi_t - \pi_t^e = -\alpha(u_t - u_n) \quad (\text{Eq. 8.9})$$

If expected inflation is well approximated by last year's inflation:

$$\pi_t - \pi_{t-1} = -\alpha(u_t - u_n) \quad (\text{Eq. 8.10})$$

NAIRU (Non-Accelerating Inflation Rate of Unemployment)

Equation 8.10 gives us two key insights: (1) The change in inflation depends on the gap between actual and natural unemployment. If $u > u_n$: inflation decreases. If $u < u_n$: inflation increases. (2) The natural rate is the unemployment rate that keeps inflation constant. This is why u_n is also called the NAIRU. From Eq. 8.7: $u_n = 3.0\%/0.5 = 6\%$ for the US (1970-2014 average).

Non-accelerating inflation rate of unemployment (NAIRU)

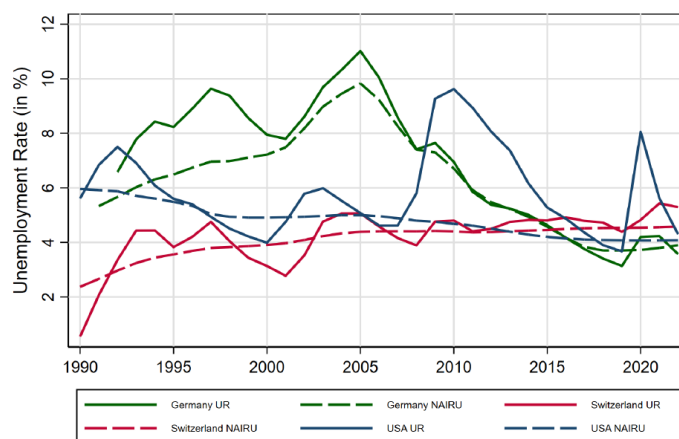
- ▶ Let us look again at $\pi_t = \pi_t^e + (\mu + z) - \alpha u_t$
- ▶ Milton Friedman and Edmund Phelps pointed out in the late 1960s that wage setters did not permanently underestimate inflation.
So, if $\pi_t = \pi_t^e$, we get $\alpha u_t = \mu + z$.
- ▶ If we solve this for u_t , we get the natural rate of unemployment:
 $u = u_n = (\mu + z)/\alpha$
- ▶ If we plug $\alpha u_n = (\mu + z)$ into the modified Phillips curve, we get:
 $\pi_t - \pi_t^e = -\alpha(u_t - u_n)$
- ▶ The Phillips curve is the relationship between the change in the inflation rate and the deviation of the unemployment rate from the natural rate of unemployment.
- ▶ That is why u_n is also known as *non-accelerating inflation rate of unemployment (NAIRU)*.

Figure: NAIRU derivation - the Phillips curve relates inflation changes to the deviation of u from u_n

NAIRU in Practice

The natural unemployment rate (NAIRU) in practice

- ▶ Medium-run tendency of the unemployment rate towards the NAIRU:



Source: OECD Economic Outlook 109, May 2021

- ▶ What economic policy conclusions follow from this?

Figure: Unemployment rates and NAIRU estimates for Germany, Switzerland, and USA (1990-2020, OECD data)

The NAIRU varies across countries and over time. Germany's NAIRU decreased substantially after the Hartz reforms. Switzerland's NAIRU has been consistently low. The US NAIRU decreased from the 1980s onwards.

WARNINGS AND VARIATIONS

8-4 Variations in the Natural Rate Across Countries

- The natural rate depends on μ , z , and α , which differ across countries
- Euro area: average unemployment around 9% since 1990, reflecting high natural rate
- Labor-market rigidities: generous unemployment insurance, high employment protection, minimum wages, extension agreements for collective bargaining
- But the picture is complex: Denmark and the Netherlands have generous social protection yet low unemployment (efficient design matters)

Variations in the Natural Rate Over Time

- $m + z$ is not constant: monopoly power, costs, bargaining structures, and benefits change over time
- US: natural rate rose from the 1950s to 1980s (from ~4.5% to ~7.3%), then declined to ~5% by 2007
- Europe: natural rate increased dramatically since the 1960s due to adverse shocks and institutional rigidities

High Inflation and the Phillips Curve

- When inflation is high, it becomes more variable and persistent
- Workers and firms shorten wage contracts and increase **wage indexation** (automatic adjustment to inflation)
- With wage indexation (proportion λ of contracts indexed):

$$\pi_t - \pi_{t-1} = -\frac{\alpha}{1-\lambda}(u_t - u_n) \quad (\text{Eq. 8.11, rearranged})$$

- Higher λ amplifies the effect of unemployment on inflation
- As λ approaches 1, small changes in unemployment cause huge inflation swings, and the Phillips curve relation breaks down

Deflation and the Phillips Curve

- During the Great Depression (1930s), high unemployment led to surprisingly little deflation
- When inflation is close to zero, workers resist nominal wage cuts (downward nominal wage rigidity)
- This means the Phillips curve may be weaker or disappear at very low inflation
- This is called **money illusion**: workers care about nominal wages, not just real wages
- Relevant today: many countries have both low inflation and high unemployment

IS-LM MODEL EXTENSIONS (LECTURE)

From Short Run to Medium Run

The IS-LM model from lecture 4 is suitable for the short run. In the medium run, we must consider:

- Resource constraints (labor market): equilibrium Y_n and u_n
- Inflation: prices change

We therefore extend the IS-LM model with: (1) distinction between nominal and real interest rates, (2) risk premiums, and (3) the Phillips curve (PC curve) -- the PC curve follows in lecture 7.

Real Interest Rate: The Fisher Equation

In the short-run IS-LM model, there was only one interest rate. With inflation, we must distinguish nominal and real interest rates.

$$\text{Fisher equation: } i_t \approx r_t + \pi_{t+1}^e$$

$$\text{where } \pi_{t+1}^e = \frac{P_{t+1}^e - P_t}{P_t}$$

- i : nominal interest rate (in terms of currency)
- r : real interest rate (in terms of goods)
- The nominal interest rate is the sum of the real interest rate and expected inflation
- Exact formula: $(1 + i_t) = (1 + r_t) * P_{t+1}^e / P_t$

The real interest rate in Switzerland

- From Swiss data on nominal interest rates and realized inflation rates, real interest rates (ex post) can be calculated:

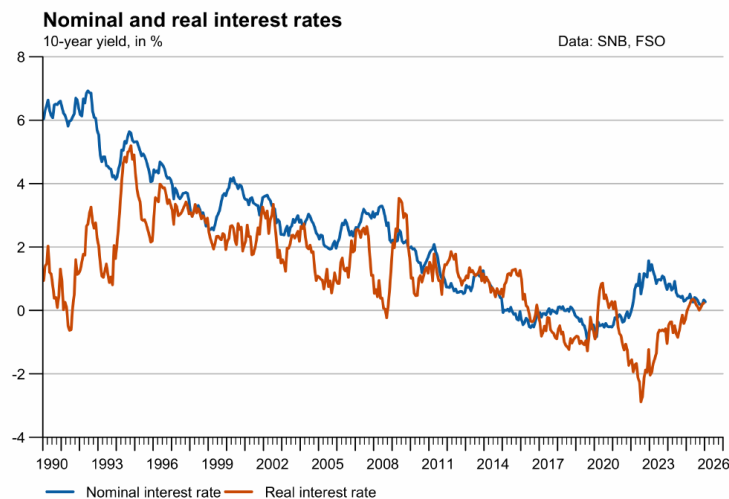


Figure: Swiss nominal and real interest rates (10-year yield, 1980-2026, Data: SNB, FSO)

Monetary Policy: The Taylor Rule

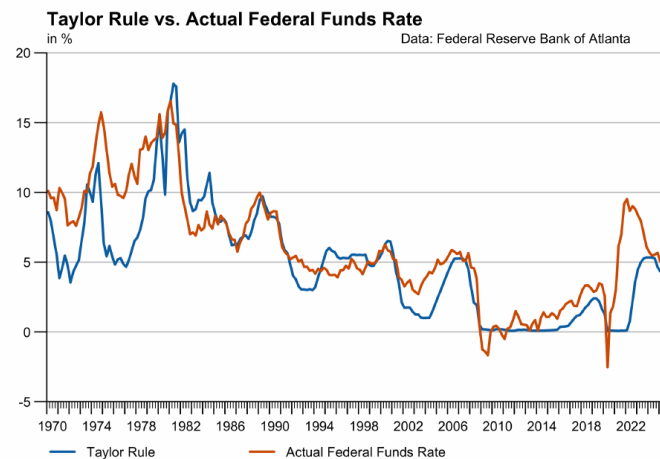
The central bank determines the real interest rate r indirectly via the nominal interest rate i . One approach for how central banks set rates:

$$i_t = r + \pi_t + a(\pi_t - \pi^*) + b(Y_t - Y_{n,t})$$

- The central bank raises i when inflation exceeds the target π^* (weight a)
- The central bank raises i when GDP exceeds potential GDP Y_n (weight b)
- Parameters a and b reflect how much value the central bank places on price stability vs economic growth
- Also known as the Orphanides rule in Europe

Taylor rule in practice

- Originally presented in 1993 by John Taylor, modified over the years:



- What are the advantages of this predictability of monetary policy?

Figure: Taylor Rule vs Actual Federal Funds Rate (1970-2022, Data: Federal Reserve Bank of Atlanta)

Risk Premium

Many bonds are risky. Buyers demand a risk premium. Let i be the safe rate, x the risk premium, and p the probability of default.

$$(1 + i) = (1 - p)(1 + i + x) + p \cdot 0$$

$$\Rightarrow x = \frac{(1 + i) \cdot p}{1 - p} \approx p \quad (\text{for small } p)$$

- The risk premium is determined by the probability of default and the degree of risk aversion
- Credit rating agencies assess default risk: AAA is safest, BBB is barely investment-grade

Extension of the IS-LM model: risk premium in the data

- The default risk of securities (*assets*) is assessed by rating agencies: AAA is the best rating, BBB is barely investment-grade.

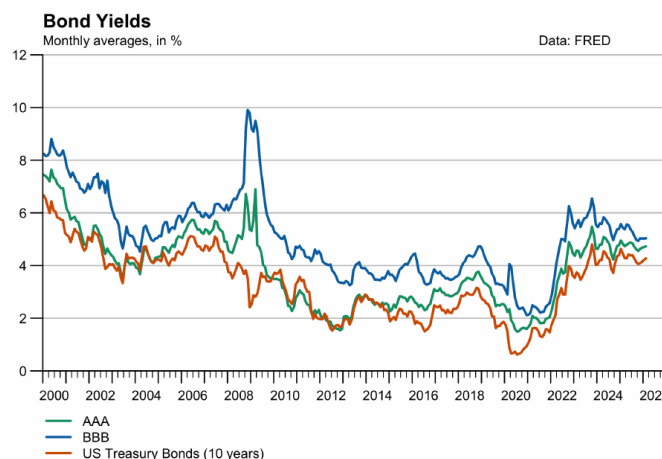


Figure: Bond yields by rating - AAA vs BBB vs US Treasuries (2000-2026, Data: FRED)

Extended IS-LM Model

Combining real interest rates and risk premiums, the IS-LM equilibrium becomes:

$$\text{IS curve: } Y = C(Y - T) + I(Y, r + x) + G$$

$$\text{LM curve: } r = r_0 = i_0 - \pi^e$$

- Investments depend on the **real interest rate plus risk premium** ($r + x$)
- The central bank controls the real interest rate through the nominal rate and inflation expectations
- The real interest rate falls when the central bank reduces i at given inflation expectations

Extended IS-LM: What Changes?

Compared to the basic IS-LM model: (1) The IS curve now uses the real interest rate + risk premium instead of the nominal rate. An increase in x shifts the IS curve left (reduces investment and output). (2) The LM curve is now in terms of the real rate: $r = i - \pi^e$. (3) The central bank must consider inflation expectations when setting the nominal rate. In lecture 7, the Phillips curve (PC curve) will be added to complete the IS-LM-PC model.

SUMMARY OF KEY EQUATIONS AND TERMS

Key Equations -- Chapter 8

$$(8.1) \quad P = P^e(1 + \mu)(1 - \alpha u + z) \quad \text{Price-wage relation}$$

$$(8.3) \quad \pi_t = \pi_t^e + (\mu + z) - \alpha u_t \quad \text{Core Phillips curve}$$

$$(8.4) \quad \pi_t = \bar{\pi} + (\mu + z) - \alpha u_t \quad \text{Original PC } (\theta = 0)$$

$$(8.5) \quad \pi_t^e = (1 - \theta)\bar{\pi} + \theta\pi_{t-1} \quad \text{Expectation formation}$$

$$(8.6) \quad \pi_t - \pi_{t-1} = (\mu + z) - \alpha u_t \quad \text{Modified PC } (\theta = 1)$$

$$(8.8) \quad u_n = \frac{\mu + z}{\alpha} \quad \text{Natural rate of unemployment}$$

$$(8.10) \quad \pi_t - \pi_{t-1} = -\alpha(u_t - u_n) \quad \text{PC in terms of } u_n$$

Key Equations -- Lecture 6 (Quantity Theory & IS-LM Extensions)

$$M \cdot V = P \cdot Y \quad \text{Quantity equation (income version)}$$

$$\% \Delta M = \% \Delta P \quad (\text{if } V \text{ constant and } M \text{ neutral})$$

$$i_t \approx r_t + \pi_{t+1}^e \quad \text{Fisher equation}$$

$$i_t = r + \pi_t + a(\pi_t - \pi^*) + b(Y_t - Y_{n,t}) \quad \text{Taylor rule}$$

$$x \approx p \quad \text{Risk premium} \approx \text{default probability}$$

$$Y = C(Y - T) + I(Y, r + x) + G \quad \text{Extended IS}$$

$$r = i_0 - \pi^e \quad \text{Extended LM}$$

Phillips Curve Evolution Summary

Period	Theta	Relation	Implication
Pre-1970	0	π vs u (level)	Stable trade-off
Post-1970	1	$\Delta \pi$ vs u	No permanent trade-off
High infl.	1 + indexation	Amplified	PC breaks down
Deflation	< 1	Weakened	Wage rigidity

Key Terms

- **Phillips curve (original)**: Negative relation between inflation level and unemployment (pre-1970)
- **Modified (expectations-augmented) Phillips curve**: Relation between change in inflation and unemployment (post-1970)
- **Accelerationist Phillips curve**: Low u leads to increasing (accelerating) inflation
- **NAIRU**: Non-accelerating inflation rate of unemployment; u_n where inflation is stable
- **Natural rate of unemployment (u_n)**: $u_n = (\mu + z) / \alpha$; depends on markup, institutions, wage sensitivity
- **Stagflation**: Simultaneous high inflation and high unemployment (1970s)
- **Inflation expectations (π^e)**: What wage setters expect inflation to be; determines actual inflation
- **Theta (expectation parameter)**: Weight on last year's inflation in expectations; rose from 0 to 1 over time
- **Wage indexation (λ)**: Proportion of contracts with automatic wage adjustment to inflation
- **Money illusion**: Workers care about nominal wages, resist nominal cuts even when real wages unchanged
- **Quantity theory of money**: $M \cdot V = P \cdot Y$; inflation is a monetary phenomenon (Friedman)
- **Velocity of money (V)**: How often each unit of money changes hands; assumed constant in QT
- **Seigniorage**: Government revenue from creating money
- **Shrinkflation**: Reducing product size while keeping price constant
- **Fisher equation**: $i = r + \pi^e$; nominal rate = real rate + expected inflation
- **Taylor rule**: $i = r + \pi + a(\pi - \pi^*) + b(Y - Y_n)$; central bank interest rate rule
- **Risk premium (x)**: Extra return demanded for bearing default risk; $x \approx p$
- **Labor-market rigidities**: Institutions that raise the natural rate: benefits, employment protection, min wages
- **Extension agreements**: Contracts between a subset of firms/unions extended to all firms in sector
- **Downward nominal wage rigidity**: Workers resist nominal wage cuts; weakens Phillips curve near zero inflation